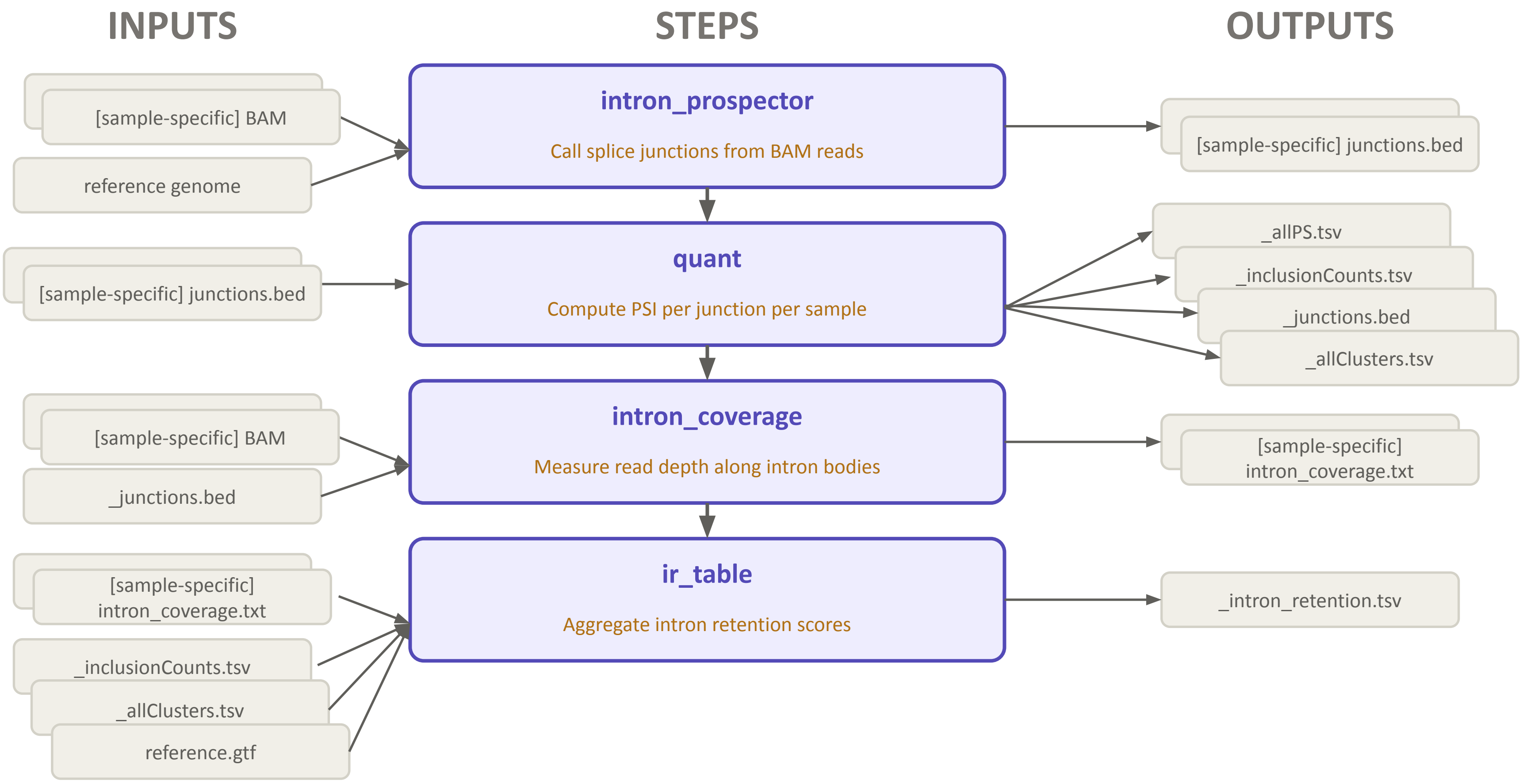
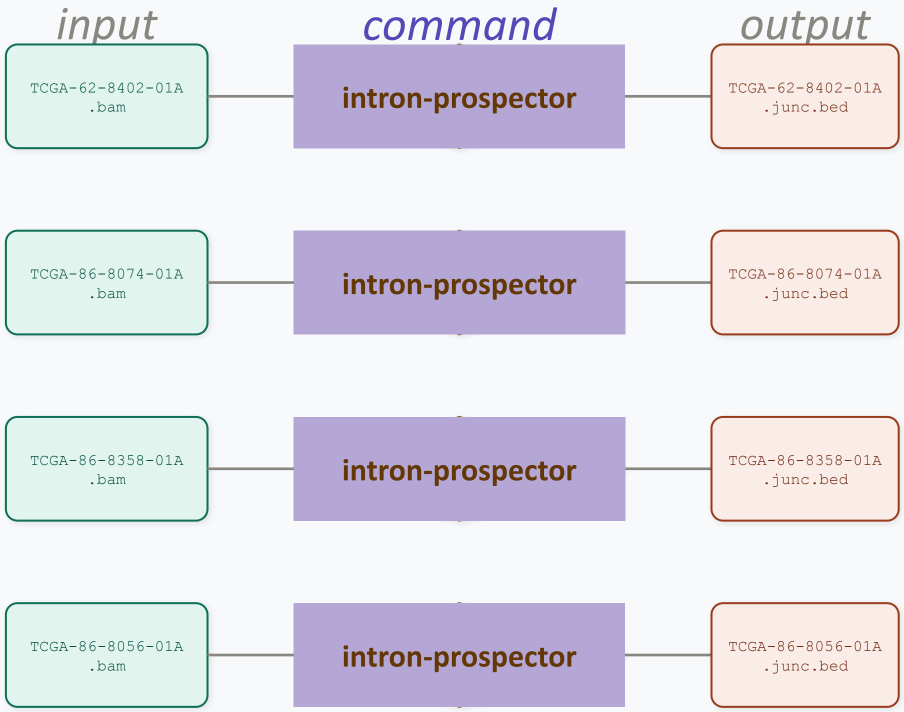
Splicedice workflow to generate percent-spliced and intron-retention data

Splicedice workflow overview



Step 1 - intron-prospector identifies junctions + coverage



each sample processed independently (parallelizable)

example input

TCGA-62-8402-01A.bam (truncated columns)

QNAME	FLAG	RNAME	POS	MAPQ	
UNC12-SN629:264:C1WVCACXX:1:1207:11678:20197	419	chr1	10474	3	...
UNC12-SN629:264:C1WVCACXX:1:1307:17161:15191	345	chr1	10534	0	...
UNC12-SN629:264:C1WVCACXX:1:1304:12112:76004	345	chr1	10569	1	...
UNC12-SN629:264:C1WVCACXX:1:2303:2554:196644	409	chr1	10570	0	...

example output

TCGA-62-8402-01A.junc.bed

chrom	start	end	name	score	strand
chr1	14829	15795	sj0_GT/AG	3	-
chr1	15038	15795	sj3_GT/AG	56	-
chr1	15947	16606	sj4_GT/AG	15	-
chr1	16765	16857	sj6_GT/AG	25	-
chr1	17055	17232	sj7_GT/AG	157	-
chr1	17368	17605	sj8_GT/AG	81	-

Step 2 - Quantify percent spliced per junction per sample

input

bed_manifest.txt

paths to per-sample .junc.bed files

```
T8074  /data/T8074.junc.bed
T8402  /data/T8402.junc.bed
T8358  /data/T8358.junc.bed
T8056  /data/T8056.junc.bed
```

command

splicedice quant

outputs

_junctions.bed

all unique splice junctions across all samples

```
chrom  start  end    name                  score  strand
chr1   14829  15795  chr1:14830-15795:-    0      -
chr1   15038  15795  chr1:15039-15795:-    0      -
chr1   15947  16606  chr1:15948-16606:-    0      -
chr1   16310  16606  chr1:16311-16606:-    0      -
chr1   16765  16857  chr1:16766-16857:-    0      -
chr1   17055  17232  chr1:17056-17232:-    0      -
```

_inclusionCounts.tsv

junction read counts per sample

```
cluster               T8074  T8402
chr1:14830-15795:-        0      0
chr1:15039-15795:-       47     55
chr1:15948-16606:-        6      8
chr1:16311-16606:-        0      2
chr1:16766-16857:-       41     13
chr1:17056-17232:-       89     73
```

_allPS.tsv

percent-spliced value per junction per sample

```
cluster             T8074  T8402  T8358  T8056
chr1:14830-15795:-  0.000  0.000  0.000  0.000
chr1:15039-15795:-  1.000  1.000  1.000  1.000
chr1:15948-16606:-  1.000  0.800  0.778  1.000
chr1:16311-16606:-  0.000  0.200  0.222  0.000
```

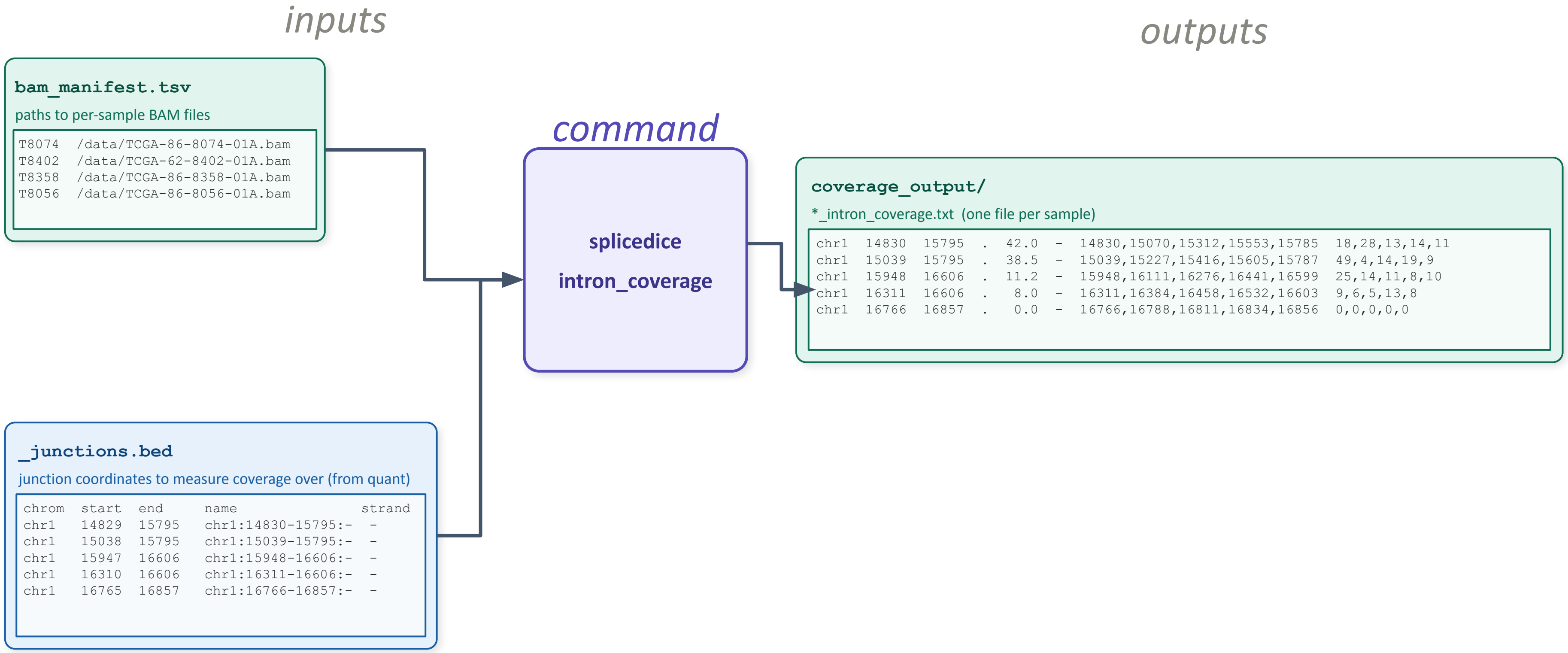
_allClusters.tsv

mutually exclusive junction pairs

```
chr1:14830-15795:-   chr1:15039-15795:-
chr1:15039-15795:-   chr1:14830-15795:-
chr1:15948-16606:-   chr1:16311-16606:-,chr1:16766-16857:-
chr1:16311-16606:-   chr1:15948-16606:-,chr1:16766-16857:-
chr1:16766-16857:-   chr1:15948-16606:-,chr1:16311-16606:-
chr1:17056-17232:-   chr1:17056-17605:-,chr1:17056-17914:-
```

Here and elsewhere I've shortened TCGA ids for display purposes, e.g. from TCGA-86-8074-01A to T8074. The original data retains the full id.

Step 3 - Calculate intron coverage



Step 4 - Generate inclusion count table

inputs

_allClusters.tsv

mutually exclusive junction pairs (from quant)

```
chr1:14830-15795:-   chr1:15039-15795:-
chr1:15039-15795:-   chr1:14830-15795:-
chr1:15948-16606:-   chr1:16311-16606:-,chr1:16766-16857:-
chr1:16311-16606:-   chr1:15948-16606:-,chr1:16766-16857:-
```

_inclusionCounts.tsv

junction read counts per sample (from quant)

cluster	T8074	T8402	T8358	T8056
chr1:14830-15795:-	0	0	0	0
chr1:15039-15795:-	47	55	52	47
chr1:15948-16606:-	6	8	7	11
chr1:16311-16606:-	0	2	2	0

annotation.gtf

junction whitelist filter — skipped with --allJunctions

```
chr1  HAVANA  gene        11121  24894  .  +
chr1  HAVANA  transcript  11121  14413  .  +
chr1  HAVANA  exon        11121  11211  .  +
chr1  HAVANA  exon        12010  12227  .  +
```

coverage_output/

*_intron_coverage.txt (one file per sample, from intron_coverage)

```
chr1  14830  15795  .  42.0  -  14830,15070,15312,15553,15785  18,28,13,14,11
chr1  15039  15795  .  38.5  -  15039,15227,15416,15605,15787  49,4,14,19,9
chr1  15948  16606  .  11.2  -  15948,16111,16276,16441,16599  25,14,11,8,10
chr1  16311  16606  .   8.0  -  16311,16384,16458,16532,16603  9,6,5,13,8
```

command

splicedice ir_table

outputs

_intron_retention.tsv

IR value per junction per sample

Junction	T8074	T8402	T8358	T8056
chr10:1000869-1000947:+	0.000	0.000	0.000	0.019
chr10:100190165-100190298:+	1.000	nan	1.000	1.000
chr10:100348347-100352365:+	0.000	0.000	0.000	0.014
chr10:100356765-100360733:+	0.003	0.005	0.000	0.006
chr10:100374013-100380982:+	0.029	0.000	0.000	0.067
chr10:100374013-100381215:+	0.000	0.000	0.000	0.000

_intron_retention_RSD.tsv

optional — produced only with --makeRSDtable flag

Junction	T8074	T8402	T8358	T8056
chr10:1000869-1000947:+	2.000	1.225	2.000	0.571
chr10:100190165-100190298:+	0.837	nan	0.333	0.848
chr10:100348347-100352365:+	1.848	nan	nan	0.935
chr10:100356765-100360733:+	1.597	0.816	nan	1.673

from quant

from intron_coverage